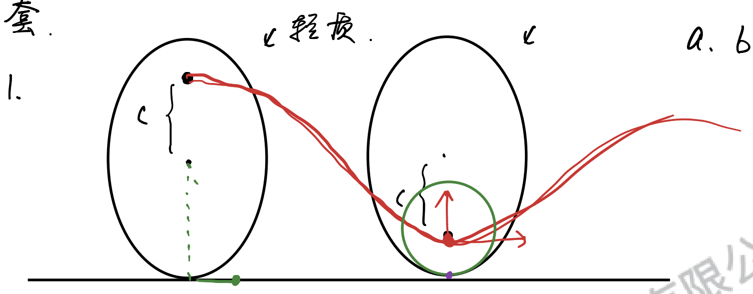
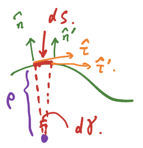
33套.

1.



a. b.



$\rho = \frac{ds}{d\gamma}$.
↑
$\hat n$转角
$= \hat\tau$转角

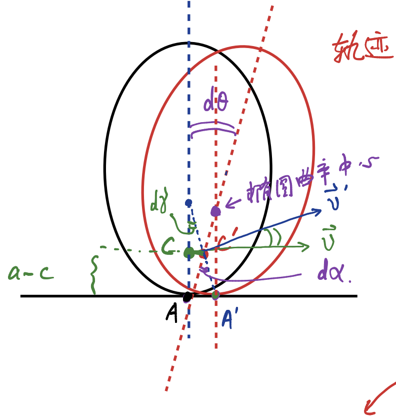
(1). 能量: $\frac{1}{2} m \cdot v_c^2 = mg \cdot 2c \Rightarrow v_c^2 = 4gc$.

法一: $\boxed{\text{先暴力求轨迹}}$ 再.求轨迹曲率半径. ρ.

← 难 ⟷ 凉凉

$\Rightarrow a_c = \frac{v_c^2}{\rho} \Rightarrow$ ✓.

法二: 先求此点之轨迹之曲率半径. $\Rightarrow \rho. \Rightarrow a_c = \frac{v_c^2}{\rho}$



轨迹. → $\rho = \frac{ds}{d\gamma} = \frac{\overline{cc'}}{d\gamma}$

$cc' = (a-c) \cdot d\theta$.

$$\boxed{d\gamma = d\alpha - d\theta}$$

纯滚约束:

地上滚过的距离

= 椭圆上.滚过的距离.

$(a-c) \cdot d\alpha$

$=$

$\rho_{椭} \cdot d\theta = \frac{b^2}{a} \cdot d\theta$.

$$\Rightarrow d\gamma = \frac{b^2}{a(a-c)} \cdot d\theta - d\theta = \frac{ac - c^2}{a^2 - ac} \cdot d\theta = \frac{c}{a} \cdot d\theta$$

$$\Rightarrow \rho = \frac{(a-c) \cdot d\theta}{\frac{c}{a} \cdot d\theta} = \frac{a(a-c)}{c}.$$ ← 轨迹曲率半径.

$$\Rightarrow a_c = \frac{v_c^2}{\rho} = \cdots = \frac{4g \cdot (a^2 - b^2)}{a(a - \sqrt{a^2 - b^2})}$$

法三. 此时.等效为.一个半径为. $\rho_{椭} = \frac{b^2}{a}$ 的圆

与地面相接触

最低点 ⇒ 角加速度为0 ⇒

⇒ $\overset{O'}{\text{圆心}}$ 加速度为0.

⇒ 质点相对圆心(椭圆曲率中心)的加速度

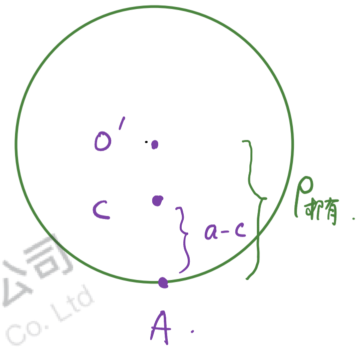
即 相对地 的加速度.

$V_c = 2\sqrt{gc} \overset{又}{=} (a-c) \cdot \omega$.

$V_{O'} = \rho_{椭} \cdot \omega = \frac{b^2}{a} \cdot \omega$.

⇒ C 相对O'的速度为:

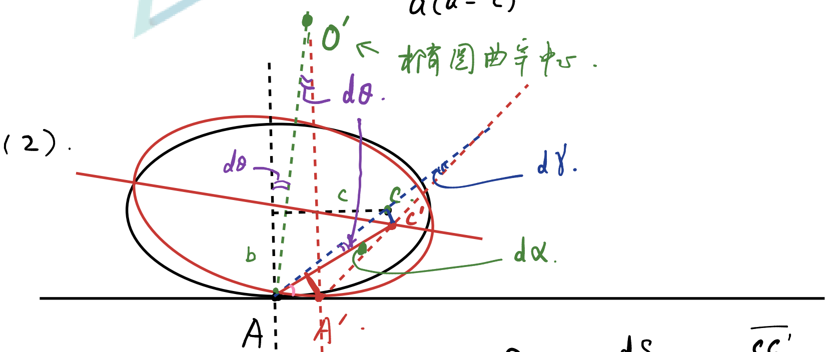
$$V_{相} = ((a-c) - \frac{b^2}{a})\omega = ((a-c) - \frac{b^2}{a}) \cdot \frac{2\sqrt{gc}}{a-c}$$



$$\Rightarrow a_{相} = \frac{(1 - \frac{b^2}{a(a-c)})^2 \cdot 4gc}{\frac{b^2}{a} - (a-c)} = \frac{\frac{(a^2 - b^2 + ac)^{\cancel{2}}}{a^2 (a-c)^2} \cdot 4gc \cdot \cancel{a}}{\cancel{b^2 - a^2 + ac}}$$

$$= \frac{c(a-c) \cdot 4gc}{a(a-c)^2} = \frac{4g(a^2 - b^2)}{a(a - \sqrt{a^2 - b^2})}.$$

(2).



$\rho_{轨} = \frac{ds}{d\gamma} = \frac{\overline{cc'}}{d\gamma}$.

$d\alpha = d\theta + d\gamma \Rightarrow d\gamma = d\alpha - d\theta$

纯滚 ⇒ 地上滚过的距离 = 椭圆上滚过的距离.

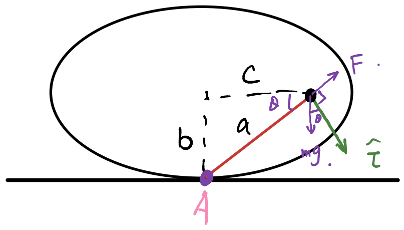
$\frac{a^2}{b} \cdot d\theta = a \cdot d\alpha \cdot \frac{a}{b} \Rightarrow d\alpha = d\theta$.

$\Rightarrow d\gamma = 0 \Rightarrow \rho_{轨} \to \infty. \Rightarrow \boxed{a_{法} = 0}$

$\Rightarrow \vec a = \vec a_{切}$.

A速度为0.
A加速度不一定为0.
加速度.不为0.

⇐

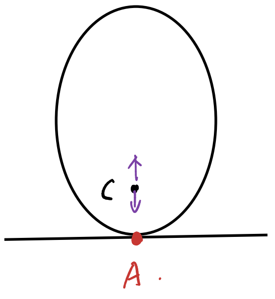


解: $\frac{1}{2} m v^2 = mg \cdot (a + c - b). \Rightarrow v^2 = 2g \cdot (a + c - b)$.

只看轻质壳: 二力杆 ⇒ 力只能沿连线.

$\Rightarrow a_{切} = a_{切} = g \cdot \cos\theta = g \cdot \frac{c}{a}$

法二: 用相对椭圆曲率中心的运动.



为什么此时.角加速度为0.

分二角度来说明：

① 轻质圆板 平衡（轻杆）.

⇒ 受力沿直径 ⇒ $\vec{F}$沿AC.

② 对C：只受竖直力

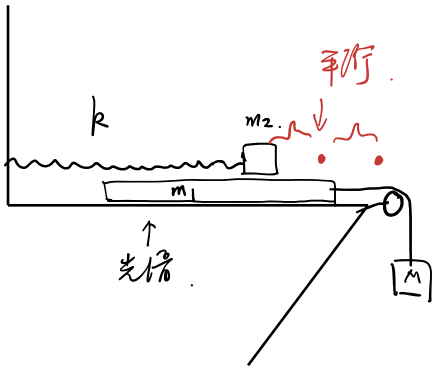
⇒ $\vec{a}_C$竖直.

③ 对A：由地面纯滚动条件：$v_{A水平}$、$a_{A水平}=0$ ⇒ $\vec{a}_A$竖直

④. 在随A的平动系中. C相对A的加速度：竖直

$\vec{a}_{相}$只沿直径. ⇒ 角加速度 $=0$.

2.



木块 木板

动、静止

释放.

牛顿力学：矢量

设断时. 伸长. x.（即移动的长度）.

~~⇒ 整体： $Mg-kx=(M+m_1+m_2)\cdot a$ ①.~~

对 M： $Mg-T=M\cdot a$.

+ 对 m_1、m_2： $T-kx=(m_1+m_2)\cdot a$

⇒ $Mg-kx=(m_1+m_2+M)\cdot a$ ✓

对 m_2： $\mu m_2\cdot g-kx=m_2\cdot a$.

$$\Rightarrow \quad \mu=\frac{kx(M+m_1)+Mm_2g}{m_2g(M+m_1+m_2)}.$$

⇒ 若一直不相对滑动. $x_{max}=2\cdot\frac{Mg}{k}$

$$\Rightarrow \quad \mu_{max}=\frac{M}{m_2}\cdot\frac{2m_1+m_2+2M}{m_1+m_2+M}.$$

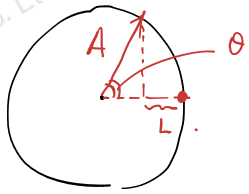
若 $\mu>\mu_{max}$ 则一直不动.

若 $\mu\le\mu_{min}=\frac{M}{m_1+m_2+M}$ 则. 刚释放就动.

(2). 并求出. 移动 L 的时间

L之前. 都无相对滑动. ⇒ 整体简谐振

$\omega=\sqrt{\frac{k}{m_1+m_2+M}}$. 振幅：$A=\frac{Mg}{k}$



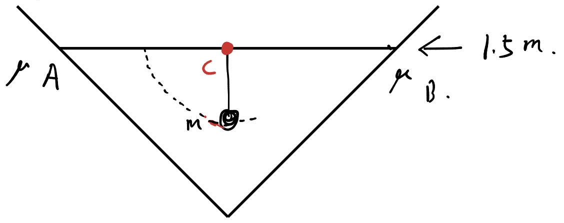
$\cos\theta=\frac{A-L}{A}$.

⇒ $\theta=\arccos\frac{A-L}{A}$

⇒ $t=\frac{\theta}{\omega}=\cdots$

已：m. μ

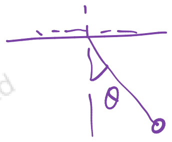
3.



刚性模型 ⇒ A.B 同时一定同失效.

全反力. + $\vec{F}_c$. ⇒ 三力. ⇒ { 矢量△. / 汇交

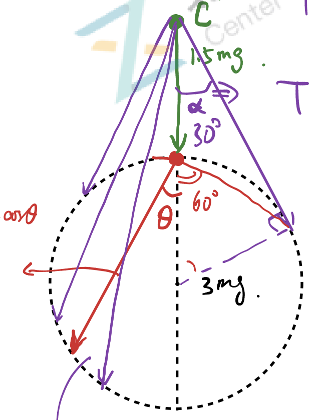
$\varphi\le\arctan\mu$. $1.5m\vec{g}+\vec{T}$.



对兔： $\frac{1}{2}mv^2=mg\cdot l\cdot\cos\theta$

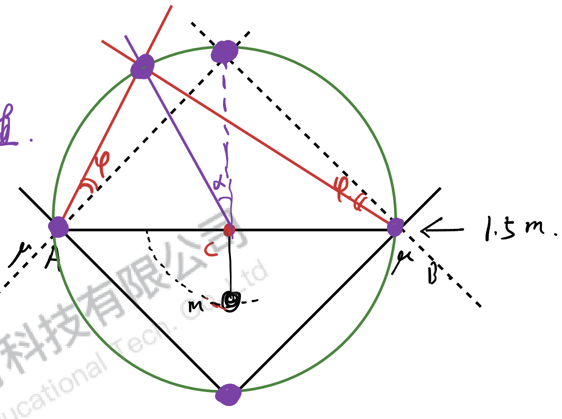
$T-mg\cdot\cos\theta=\frac{mv^2}{l}$.

$T=mg\cdot\cos\theta+2mg\cdot\cos\theta=3mg\cdot\cos\theta$.



← $\vec{T}$的存在的圆.

$\vec{F}_c$的所有可能取值.



⇒ 当$\vec{F}_c$偏离竖直

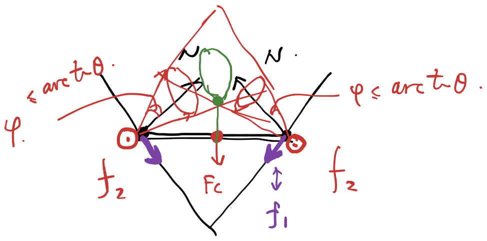
最远时.

对μ要求最高

⇒ $\vec{F}_c$在圆的切线处： $\alpha_{max}=30°$ ⇒ $\varphi_{max}=15°$

⇒ μ至少为 tan15°. $=\cdots$ ✓.

(2) ⊥纸面摆动.



对杆:

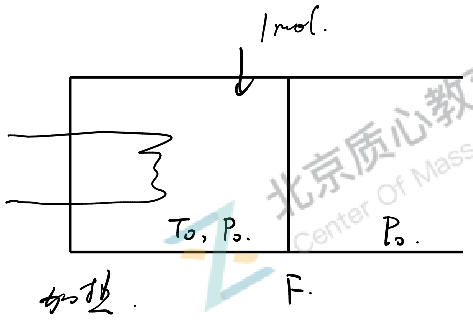
竖直方向平衡: $2\cdot\frac{\sqrt{2}}{2}\cdot N = 1.5mg + 3mg\cdot\cos\theta\cdot\cos\theta$.

水平: 垂直于纸面: $2f = 3mg\cdot\cos\theta\cdot\sin\theta$

$$\Rightarrow \mu \geq \frac{f}{N} = \frac{1.5\cdot\sin\theta\cos\theta}{\frac{1}{\sqrt{2}}\cdot(1.5+3\cdot\cos^2\theta)} = \frac{1.5\cdot\tan\theta}{\frac{1}{\sqrt{2}}\cdot(1.5(1+\tan^2\theta)+3)} = \frac{1.5}{\frac{1}{\sqrt{2}}\cdot(4.5\cdot\frac{1}{\tan\theta}+1.5\cdot\tan\theta)}$$

$\Rightarrow \mu = \frac{1}{\sqrt{6}}$. $\mu < \frac{1}{\sqrt{6}}$

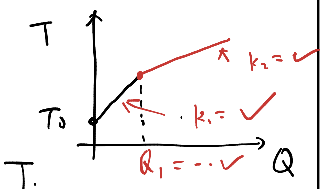
4.



1mol. $u = cT$.

F.

吸收一半之摩擦热



Q: 电热丝之热.

2过程: 第1个: 没动. ⇒ $Q \to T$.

第2个: 动.

1过程: $Q = C(T-T_0)$ ⇒ $T = \frac{Q}{c} + T_0$

当: $P_1\cdot S - P_0\cdot S = F$ 时. 动.

状态: $\frac{P_1}{T_1} = \frac{P_0}{T_0} \Rightarrow T_1 = \frac{T_0}{P_0}(P_0 + \frac{F}{S})$

⇒ $Q_1 = \cdots$ ⇒ 开始第2个过程.

⇒ 等压过程:

$\Delta Q_总 = C_p\cdot\Delta T = (C+R)\cdot\Delta T$.

⇓

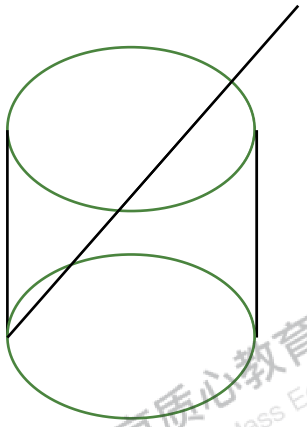
$\Delta Q_总 = \underset{\substack{\uparrow\\ 电热丝}}{\Delta Q} + \frac{1}{2}\cdot F\cdot\frac{\Delta V}{S}$ ←

由状态: $P_1\cdot V = R\cdot T \Rightarrow \Delta V = \frac{R}{P_1}\cdot\Delta T$.

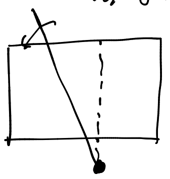
$\Rightarrow \Delta Q + \frac{1}{2}\cdot F\cdot\frac{1}{S}\cdot\frac{R}{P_1}\cdot\Delta T = (C+R)\cdot\Delta T$.

$\Rightarrow \frac{\Delta T}{\Delta Q} = \cdots = 常$ ⇒ 直线.

题3.

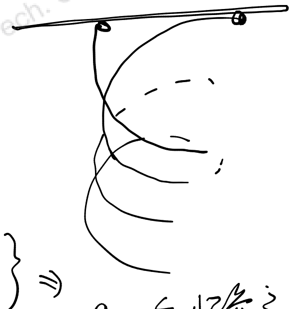


何时滑动.



何时滑.

思考: 何时滑.



⇒ ① 静摩擦之临界情况. ② 动摩擦之临界情况 } ⇒ 在2个临界之间.

与初始情况有关: { 初始动: 不平衡 ⇒ 动. ; 初不动: 平衡 ⇒ 不动.

题3. 用静摩擦约束去讨论

法一: 用摩擦锥. ⇒ 几何. ⇒ 解析.

法二: 用平衡.

$\Rightarrow \sqrt{2}N - \sqrt{2}f_1 = 1.5mg + 3mg\cdot\frac{\sqrt{3}}{4}$.

$\Rightarrow N = f_1 + (1.5 + \frac{3\sqrt{3}}{4})mg$.

$2\cdot\frac{\sqrt{2}}{2}\cdot N - 2\cdot\frac{\sqrt{2}}{2}\cdot f_1 = 1.5mg + 3mg\cdot\cos\theta\cdot\cos\theta$. ①.

析. 垂直于纸面: $2f_2 = 3mg\cdot\cos\theta\cdot\sin\theta$

要求. $\mu \geq \frac{\sqrt{f_1^2+f_2^2}}{N}$.

对μ要求最高时.

$\alpha = 30° \Rightarrow \theta = 60°$

$\Rightarrow f_2 = 1.5mg\cdot\frac{\sqrt{3}}{4}$.

$$\Rightarrow \mu \geq \frac{\sqrt{\left(\frac{3\sqrt{3}}{8}\right)^2 + f_1^2}}{f_1 + 1.5 + \frac{3\sqrt{3}}{4}} \Rightarrow \text{取极值}$$

$$\Rightarrow \mu \geq \frac{1}{\sqrt{7}}.$$

静摩擦：要求 $\mu \geq \frac{1}{\sqrt{7}}$ 即可.

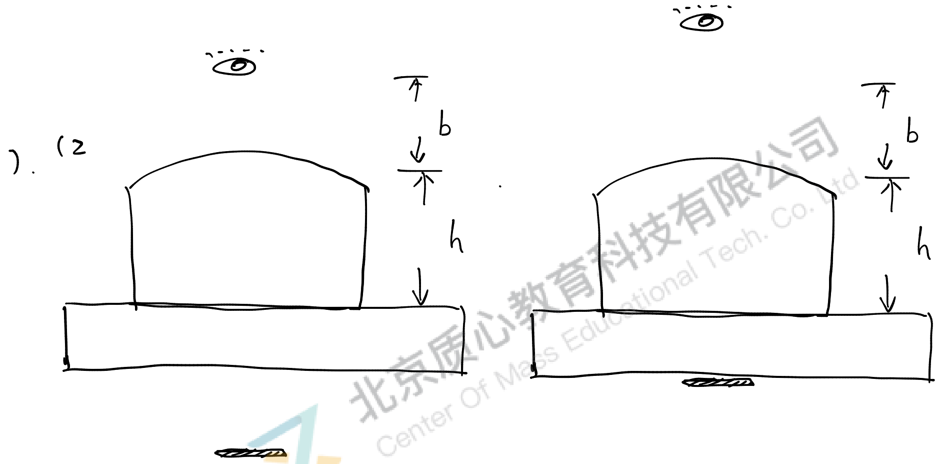
用动摩擦：要求：$\mu \geq \frac{1}{\sqrt{6}}$ 才行

当 $\frac{1}{\sqrt{7}} \leq \mu < \frac{1}{\sqrt{6}}$ 时：初态. ← 不动. ⇒ 平衡.

5题：(1). $n = \dfrac{h}{b - a - \frac{d}{n_0}}$

(2). 有2次. 俏和小广告重合.

⇒ $n = 1.3$, $R = 30.0\ \text{cm}$



小广告：①.经玻璃、视深.

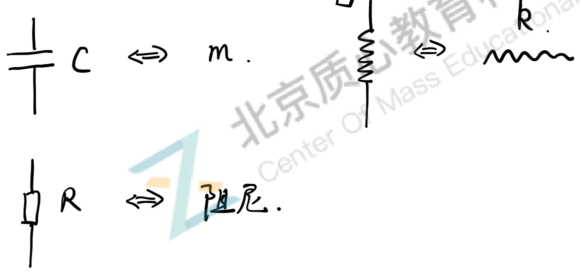
②.经.果冻/玻璃、视深.

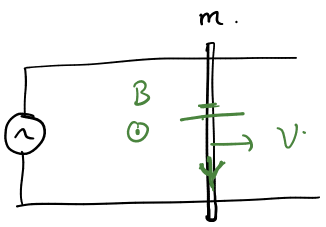
③.经.果冻上表面：单球面折射

⇒ 成像 → 重合.

眼睛：只经.果冻上表面反射成像

题6. 电磁动力学中:



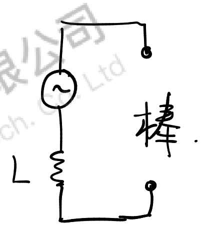


$f = -k\vec{v}$.

回路：L

对棒：

⇒



$U_1 = Bl\cdot v$.

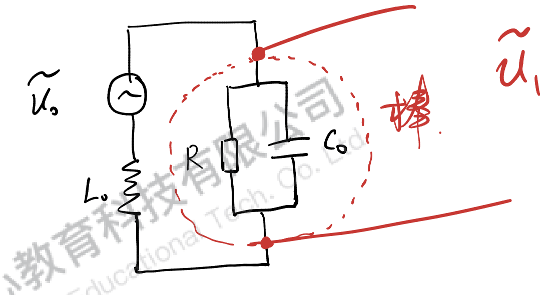
牛二：$m\cdot\frac{dv}{dt} = -Bl\cdot i - k\cdot v$.

⇒ 化为纯电学量：$m\cdot\frac{1}{Bl}\frac{du_1}{dt} = -Bl\cdot i - k\cdot\frac{u_1}{Bl}$.

$$\Rightarrow i = -\underset{\substack{\updownarrow\\ \text{电容电流}}}{\frac{m}{B^2l^2}\cdot\dot{u}_1} - \underset{\substack{\updownarrow\\ \text{电阻上电流}}}{\frac{k}{B^2l^2}\cdot u_1}.$$

⇒.

⇒. 并联. ⇒



$R = \frac{B^2l^2}{k}$, $C = \frac{m}{B^2l^2}$.

(2). $U_{(t)1} = U_0\cdot\cos(\omega t)$. 求.振幅

⇕

稳定后的情况

⇙

不考虑暂态过程.

用电学方式解. ⇒ 对棒：$\tilde{u}_1 = Bl\cdot\tilde{v}$.

电路中：$\tilde{u}_1 = \tilde{u}_0\cdot\dfrac{\frac{1}{i\omega C_0}\cdot R}{i\omega L_0 + \frac{\frac{1}{i\omega C_0}\cdot R}{\frac{1}{i\omega C_0}+R}} = \cdots$

$\tilde{v} = i\omega\cdot\tilde{x}$. ⇒ $\tilde{x} = \frac{\tilde{v}}{i\omega} = \frac{\tilde{u}_1}{i\omega\cdot Bl}$.

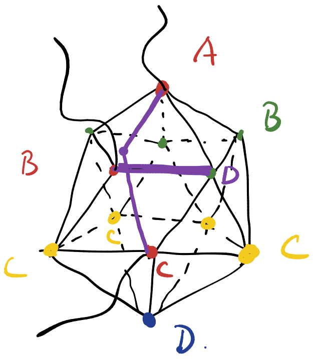
⇒ 求其模长即可.

法二：阻尼方法：把所有电学量 换为力学量

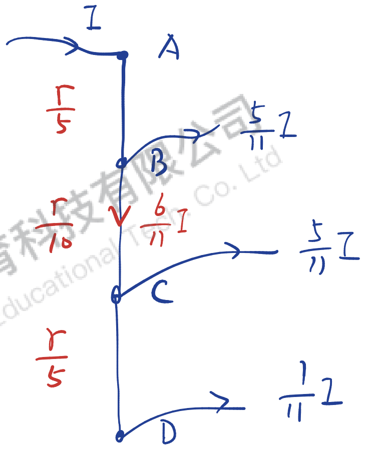
⇒ $\ddot{x} +$ [illegible] $\cdot\dot{x} +$ [illegible] $x =$ [illegible] $\cdot\cos(\omega t)$

⇒ 解特解 即可 ⇒ ✓

题7. (1) 伏安法. ①从 A 入 I. 从余下每个出 $\frac{I}{11}$.
②从 C 出 I. 从余下每个入 $\frac{I}{11}$.



⇒



$$\Rightarrow U_{AC1} = \frac{r}{5}I + \frac{r}{10}\cdot\frac{6}{11}I$$

$$= \cdots$$

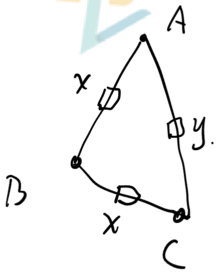
$$\Rightarrow R_{AC1} = \frac{2U_{AC1}}{\frac{12}{11}I} \Rightarrow \frac{7}{15}r.$$

(2) $R_{AC1} = R_{AC2}$ 并 $2r$.

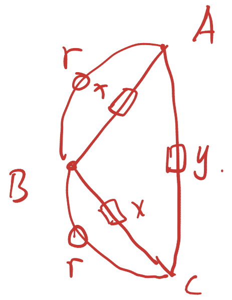
$$\Rightarrow R_{AC2} = \frac{14}{23}r.$$

(3). 去掉 AB、BC之后.

看. A、B、C 之三端网络.



$\Rightarrow R_{AC2} = 2x$ 并 y



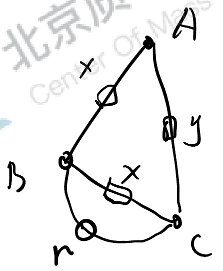
没拆任何R时.

⇒

$R_{AB} = (r并x)并[(r并x)串y]$

$$\begin{cases} x = \cdots \\ y = \cdots \end{cases}$$

(3) 问:



$\Rightarrow R_{AC3} = \cdots$ ✓